2. The parabola $y^2 = kx$ passes through the point (4,–2). Find its latus rectum and focus.

3. Find the vertex, focus, axis, directrix and the length of latus rectum of the parabola $y^2 - 8y - 8x + 24 = 0$

4. Find the co-ordinates of the focus, vertex, equation of the directrix, axis and the length of latus rectum of the parabola

   (a) $y^2 = 20x$

   (b) $x^2 = 8y$

   (c) $x^2 = -16y$

   IAEFNN

5. The average variable cost of a monthly output of x tonnes of a firm producing a valuable metal is ₹ $\frac{1}{5}x^2 - 6x + 100$. Show that the average variable cost curve is a parabola. Also find the output and the average cost at the vertex of the parabola.

6. The profit ₹y accumulated in thousand in x months is given by $y = -x^2 + 10x - 15$. Find the best time to end the project.

Exercise 3.7

Choose the correct answer

1. If m_1 and m_2 are the slopes of the pair of lines given by $ax^2 + 2hxy + by^2 = 0$, then the value of $m_1 + m_2$ is

   (a) $\frac{2h}{b}$ (b) $-\frac{2h}{b}$

   (c) $\frac{2h}{a}$ (d) $-\frac{2h}{a}$

2. The angle between the pair of straight lines $x^2 - 7xy + 4y^2 = 0$ is

   (a) $\tan^{-1}\left(\frac{1}{3}\right)$ (b) $\tan^{-1}\left(\frac{1}{2}\right)$

   (c) $\tan^{-1}\left(\frac{\sqrt{33}}{5}\right)$ (d) $\tan^{-1}\left(\frac{5}{\sqrt{33}}\right)$

3. If the lines $2x - 3y - 5 = 0$ and $3x - 4y - 7 = 0$ are the diameters of a circle, then its centre is

   (a) (–1, 1) (b) (1,1)

   (c) (1, –1) (d) (–1, –1)

4. The x–intercept of the straight line $3x + 2y - 1 = 0$ is

   (a) 3 (b) 2 (c) $\frac{1}{3}$ (d) $\frac{1}{2}$

5. The slope of the line $7x + 5y - 8 = 0$ is

   (a) $\frac{7}{5}$ (b) $-\frac{7}{5}$ (c) $\frac{5}{7}$ (d) $-\frac{5}{7}$

6. The locus of the point P which moves such that P is at equidistance from their coordinate axes is

   (a) $y = \frac{1}{x}$ (b) $y = -x$

   (c) $y = x$ (d) $y = \frac{-1}{x}$

7. The locus of the point P which moves such that P is always at equidistance from the line $x + 2y + 7 = 0$ is

   (a) $x + 2y + 2 = 0$ (b) $x - 2y + 1 = 0$

   (c) $2x - y + 2 = 0$ (d) $3x + y + 1 = 0$

8. If $kx^2 + 3xy - 2y^2 = 0$ represent a pair of lines which are perpendicular then k is equal to

   (a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) 2 (d) –2

9. (1, –2) is the centre of the circle $x^2 + y^2 + ax + by - 4 = 0$, then its radius

   (a) 3 (b) 2 (c) 4 (d) 1

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